Fundamentals of Avionics EL:

Antenna Design for Miniaturised Low Power Synthetic Aperture Radar for UAVs

Avionics EL

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INTRODUCTION

- Synthetic Aperture Radar (SAR) is a high-resolution imaging radar system used for Earth observation, environmental monitoring, and reconnaissance, operating effectively in all weather conditions.
- With the growing demand, there is a need to downscale SAR systems without compromising performance.
- A key component of SAR is the antenna, which must be compact, lightweight, and efficient for onboard satellite deployment.
- Microstrip Patch Antennas (MPAs) offer a promising solution due to their:
- Low profile and conformal shape
- Ease of fabrication and integration
- Suitability for high-frequency applications
- This project focuses on the design, simulation, and optimization of a microstrip patch antenna tailored for miniaturized SAR systems, targeting improved performance in terms of gain, bandwidth, beamwidth, and polarization.

LITERATURE SURVEY

Author(s)	Paper Title	Publication Details	Summary
Ramanuja Manavalan(2018)	Review of synthetic aperture radar frequency, polarization, and incidence angle data for mapping the inundated regions	J. Appl. Remote Sens. 12(2), 021501 (2018)	SAR is crucial for flood mapping as it works in all weather and captures surface reflections through clouds. This review compiles key factors like frequency, polarization, and geography affecting SAR-based flood detection accuracy.
Done et al. (2017)	Design and implementation of a satellite communication ground station	IEEE ISSCS Conference Proceedings, 2017	Presented a low-cost satellite ground station for LEO satellites, operating on 145 & 436 MHz bands, suitable for academic use and telemetry/data download.
Hwang (2022)	Ocean Surface Roughness from Satellite Observations and Spectrum Modeling	Journal of Physical Oceanography, AMS, 2022	Modeled ocean surface roughness using satellite-derived LPMSS data; showed wave breaking contributes to both energy dissipation and roughness; introduced a non-linear spectral function with exponent ~0.38.

LITERATURE SURVEY

Author(s)	Paper Title	Publication Details	Summary
Sun et al. (2023)	Effects of microplastics and surfactants on surface roughness of water waves	Remote Sensing of Environment or similar (assumed)	Found surfactants significantly reduce surface roughness; microplastics only affect MSS at surface coverages >5–10%, suggesting surfactants are the main contributors to MSS damping in real-world conditions.
HyperCast Team (2023)	HyperCast: Hyperspectral satellite image broadcasting with band ordering optimization	Conference or Journal on Satellite Communication (assumed)	Introduced a method for optimized spectral band transmission from satellites, allowing partial image reconstruction and efficient use of bandwidth in hyperspectral remote sensing applications.

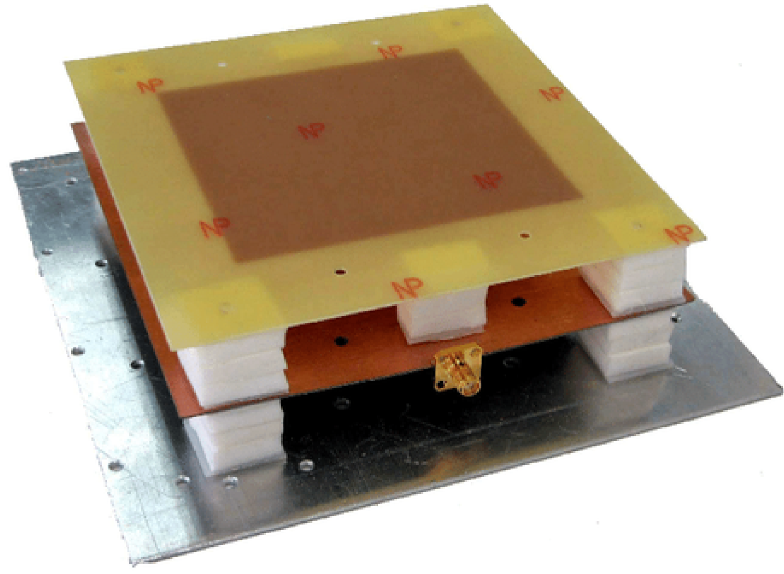
SUBSTRATES USED:

There are various types of substrates widely used in antenna applications, however, each comes with its own set of disadvantages

- Ceramics Based: Hard and brittle to process mechanically, max size limit is 4 inch by 4 inch. ✗
- Synthetic Based: Pure organic materials, soft and not good at low extreme temperature. ✗
- Semiconductor Based: Too small to be used at microwave frequencies ✗

Composite substrates are considered the most suitable option for this application. ✓

- Good permittivity (2.1 to 10)
- Less loss tangent (0.0005 to 0.0002) at 1 GHz
- Flexible in sizes, max size is 1m
- PTFE, Quartz, Honeycomb etc. have good radiation efficiency but cost for fabricating is much high
- FR4 have good strengths but have high loss tangent



The best choice for substrate depends on dielectric constant and loss tangent

- If $\varepsilon r > 10$, antenna size will be less, Bandwidth narrower and loss tangent will be less
- If $\varepsilon r < 2$, antenna will be larger, Bandwidth wider and loss tangent will be less

SUBSTRATE USED

Frequency Selection: 2 GHz (L-band)
Offers good penetration through the atmosphere and suitable resolution for ocean surface imaging.
Target operations: Ocean surface topography and oil spill detection

Based on the target operations, several substrates were used:

Substrate Material:

Rogers RO4003C

Dielectric Constant (ε_r): 3.38 – balanced bandwidth and antenna size
Loss Tangent: 0.0021 – 0.0025 – ensures minimal signal loss

SUBSTRATE USED

Substrate Material:

Duroid 5880

Dielectric Constant (εr): 2.20 – ideal for precision RF/microwave circuits

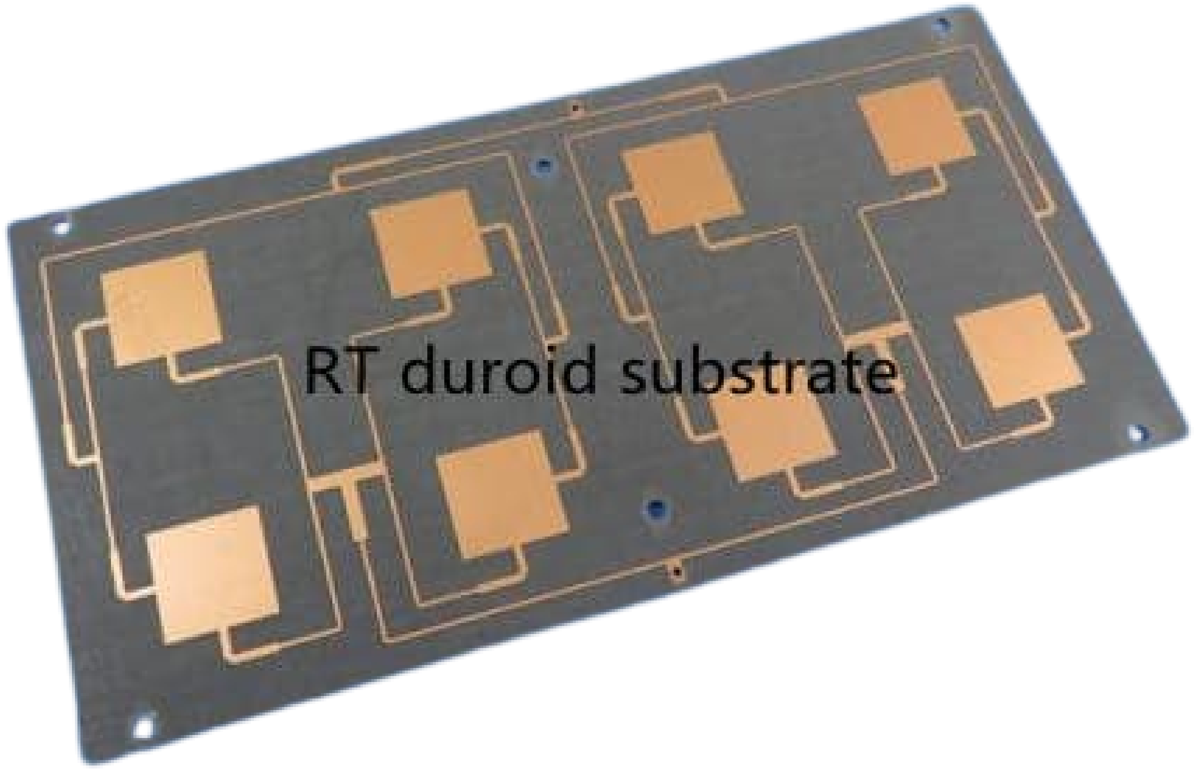
Loss Tangent: 0.0005 – 0.0007 – ensures minimal signal loss

Substrate Material:

FR-4 (Epoxy Glass)

Dielectric Constant (εr): 4.4

Loss Tangent: 0.018 – 0.020 – ensures minimal signal loss



Substrate Material:

Taconic TLY-5

Dielectric Constant (εr): 2.20 – Comparable to Duroid 5880 in low-loss performance

Loss Tangent: 0.0007 – 0.0009

Dimension Calculations

To calculate important dimensions for the Patch and the Substrate, the following Python code was utilised:

```
import math

eps_r = float(input("Enter the relative dielectric constant (εr): "))
f = float(input("Enter the operating frequency in Hz (e.g. 2e9 for 2 GHz): "))
h_mm = float(input("Enter the substrate thickness in mm (e.g. 3.2): "))
margin = float(input("Enter the margin in mm (e.g. 5): "))
# Constants
c = 3e8             # Speed of light (m/s)
h = h_mm / 1000       # Convert thickness to meters
margin_m = margin / 1000

# 1. Patch Width (W)
W = (c / (2 * f)) * math.sqrt(2 / (eps_r + 1))

# 2. Effective Dielectric Constant (ε_eff)
eps_eff = (eps_r + 1)/2 + (eps_r - 1)/2 * (1 + 12 * (h / W))-0.5

# 3. Fringing Length ΔL
delta_L = h * 0.412 * ((eps_eff + 0.3) * ((W / h) + 0.264)) / ((eps_eff - 0.258) * ((W / h) + 0.8))
# 4. Patch Length (L)
L = (c / (2 * f * math.sqrt(eps_eff))) - 2 * delta_L
# 5. A = c / (2f)
A = c / (2 * f)
# 6. C = (2 * h_mm) / π
C = (2 * h_mm) / math.pi
# 7. D = L / π * cos⁻¹(√(50/100))
D = (L / math.pi) * math.acos(math.sqrt(50 / 100))
# 8. E = √2 × D
E = math.sqrt(2) * D
# 9. Substrate Dimensions
substrate_width = 2 * W + (A-W) + 2 * margin_m
substrate_length = L + 3 * D + 2 * margin_m
# Output
print("\n📐 Microstrip Patch Antenna Parameters:")
print("Patch Width (W): {:.3f} mm".format(W * 1000))
print("Patch Length (L): {:.3f} mm".format(L * 1000))
print("Effective Dielectric Constant (ε_eff): {:.4f}".format(eps_eff))
print("Fringing Length (ΔL): {:.3f} mm".format(delta_L * 1000))
print("\n📏 Additional Parameters:")
print("A (Center-to-center λ/2): {:.3f} mm".format(A * 1000))
print("C (Inset depth): {:.3f} mm".format(C))
print("D (Inset feed shift): {:.3f} mm".format(D * 1000))
print("E (T-connector length): {:.3f} mm".format(E * 1000))
print("\n📦 Substrate Dimensions:")
print("Substrate Width: {:.3f} mm".format(substrate_width * 1000))
print("Substrate Length: {:.3f} mm".format(substrate_length * 1000))
```

Formulas Used

The thickness of the substrate (h) significantly affects the performance of the antenna, especially impedance matching and bandwidth.

- It is generally chosen within a specific range relative to the effective wavelength in the dielectric material.
- The recommended range is given by the inequality:

$$0.0065 \leq \frac{h}{\lambda_d} \leq 0.1292$$

Where:

h = substrate thickness

λd = wavelength in the dielectric medium, given by $$\lambda_d = \frac{\lambda_0}{\sqrt{\varepsilon_r}}$$

Substituting in the inequality,

$$0.0065 \times \lambda_d \leq h \leq 0.1292 \times \lambda_d$$

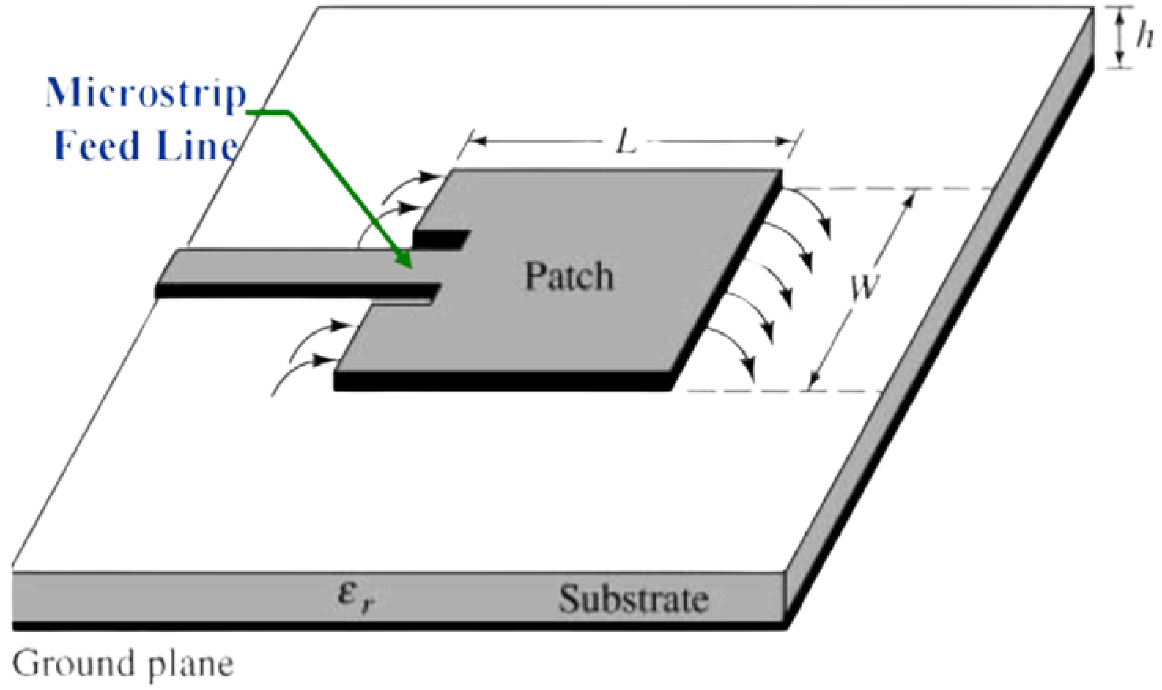
$$0.0065 \times 0.06123 \leq h \leq 0.1292 \times 0.06123$$

$$0.000398 \leq h \leq 0.00791 \text{ m} \quad \Rightarrow \quad 0.398 \text{ mm} \leq h \leq 7.91 \text{ mm}$$

- 0.39 mm ≤h <2 mm, Narrow bandwidth, Lower radiation efficiency and Poor impedance matching
- 2 mm ≤h ≤5 mm, Balanced performance, Good bandwidth and Proper matching
- 5 mm <h ≤7.9 mm, Surface wave loss and very large size

ANTENNA PATCH

WIDTH



Accurate computation of patch dimensions is essential to achieve the desired resonant frequency, impedance matching, and optimal radiation performance.

The dimensions of the antenna is given by,

$$\text{Width:}\quad W = \frac{V_0}{2f_r}\sqrt{\frac{2}{\varepsilon_r + 1}}$$

Where:

V0: Speed of light in free space (3× 10^8 m/s)

fr: Resonant frequency which is same as operating frequency in our case because antenna is resonant

εr = dielectric constant of the substrate

ANTENNA PATCH

EFFECTIVE DILECTRIC CONSTANT

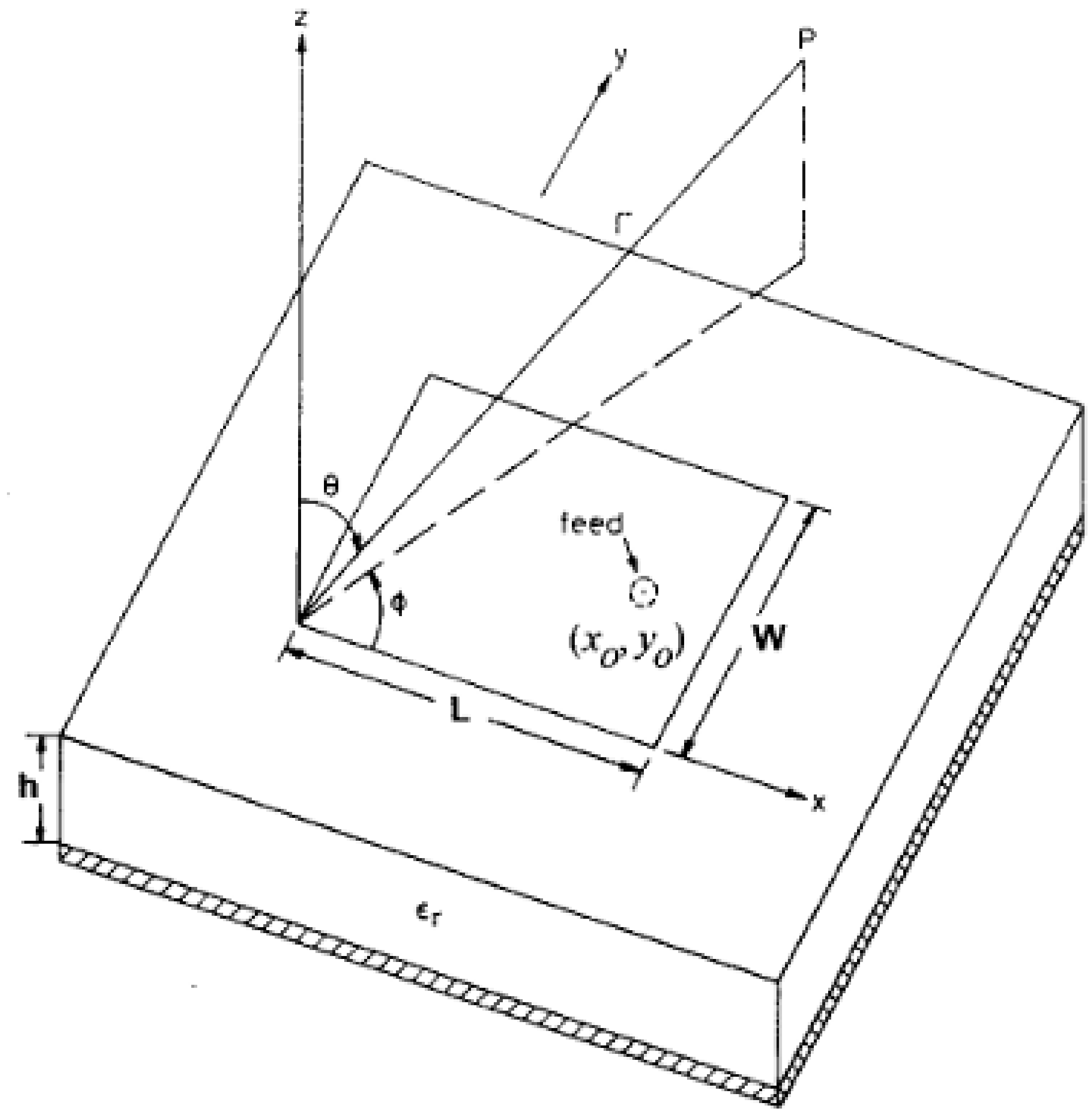
$$\varepsilon_{\text{eff}} = \frac{\varepsilon_r + 1}{2} + \frac{\varepsilon_r - 1}{2}\left(1 + 12\frac{h}{W}\right)^{-1/}$$

Where:

W is width of the patch antenna

h is the height of substrate

εr = dielectric constant of the substrate



LENGTH OF PATCH

$$L = \frac{V_0}{2 f_r \sqrt{\varepsilon_{\text{eff}}}} - 2\Delta L$$

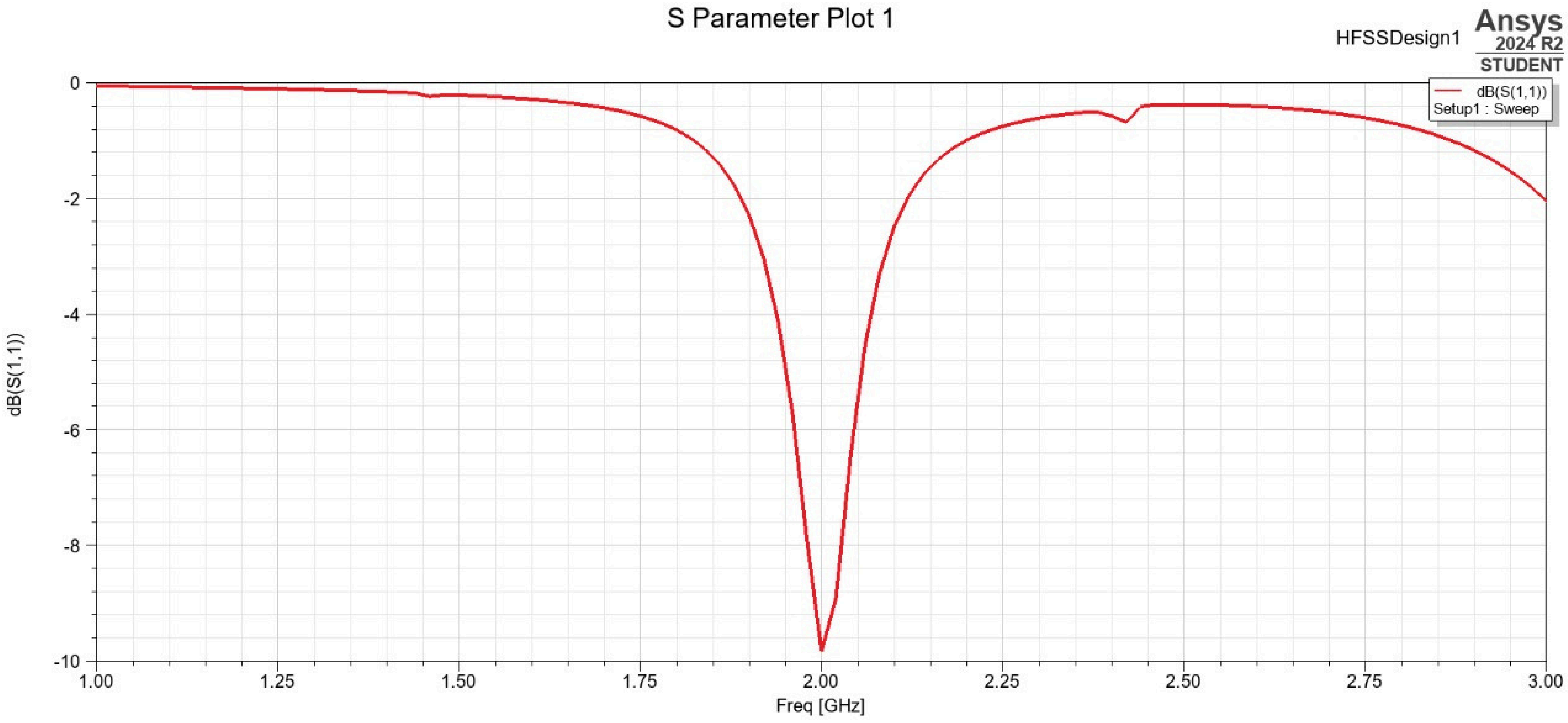
RESULT

Go, Change the World

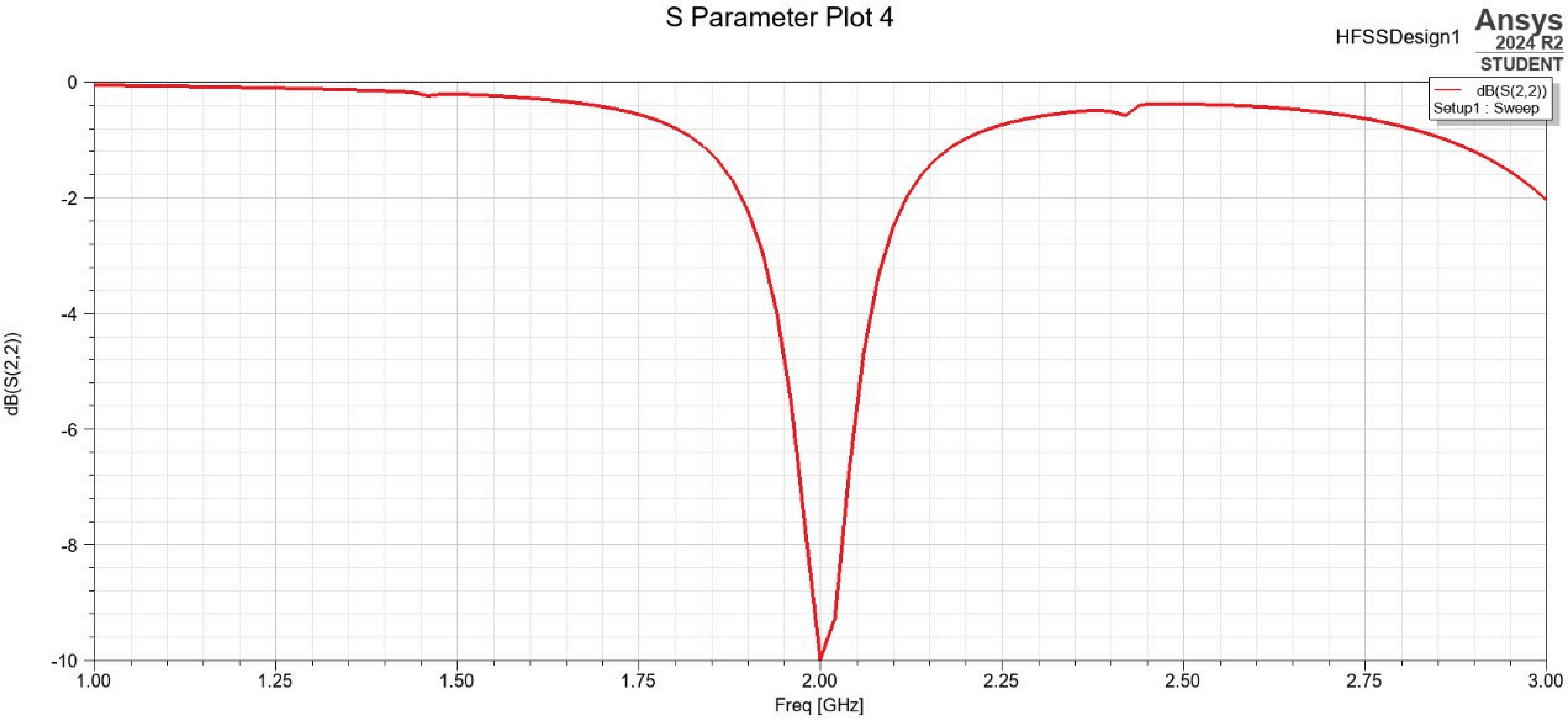
Substrate	Patch W (mm)	Patch L (mm)	Substrate W (mm)	Substrate L (mm)	εr	A (mm)	C (mm)	D (mm)
RO4003C	49.725	38.685	194.449	87.698	3.55	75.0	2.037	9.671
RT/duroid 5880	59.293	48.807	213.585	105.412	2.2	75.0	2.037	12.202
FR-4	45.227	34.41	185.453	80.218	4.5	75.0	2.037	8.603
Taconic TLY-5	59.293	48.807	213.585	105.412	2.2	75.0	2.037	12.202

RT/duroid 5880

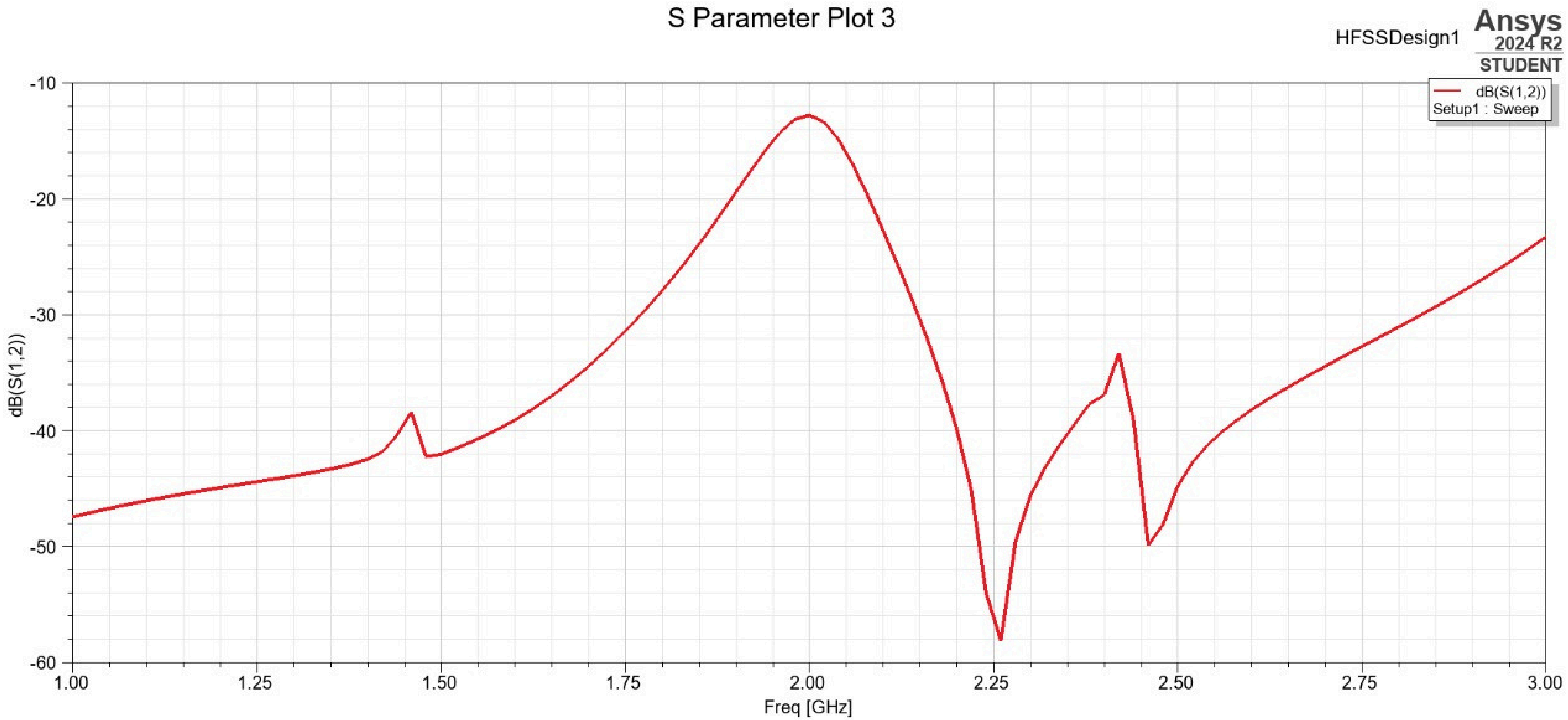
S11 Plot



S22 Plot

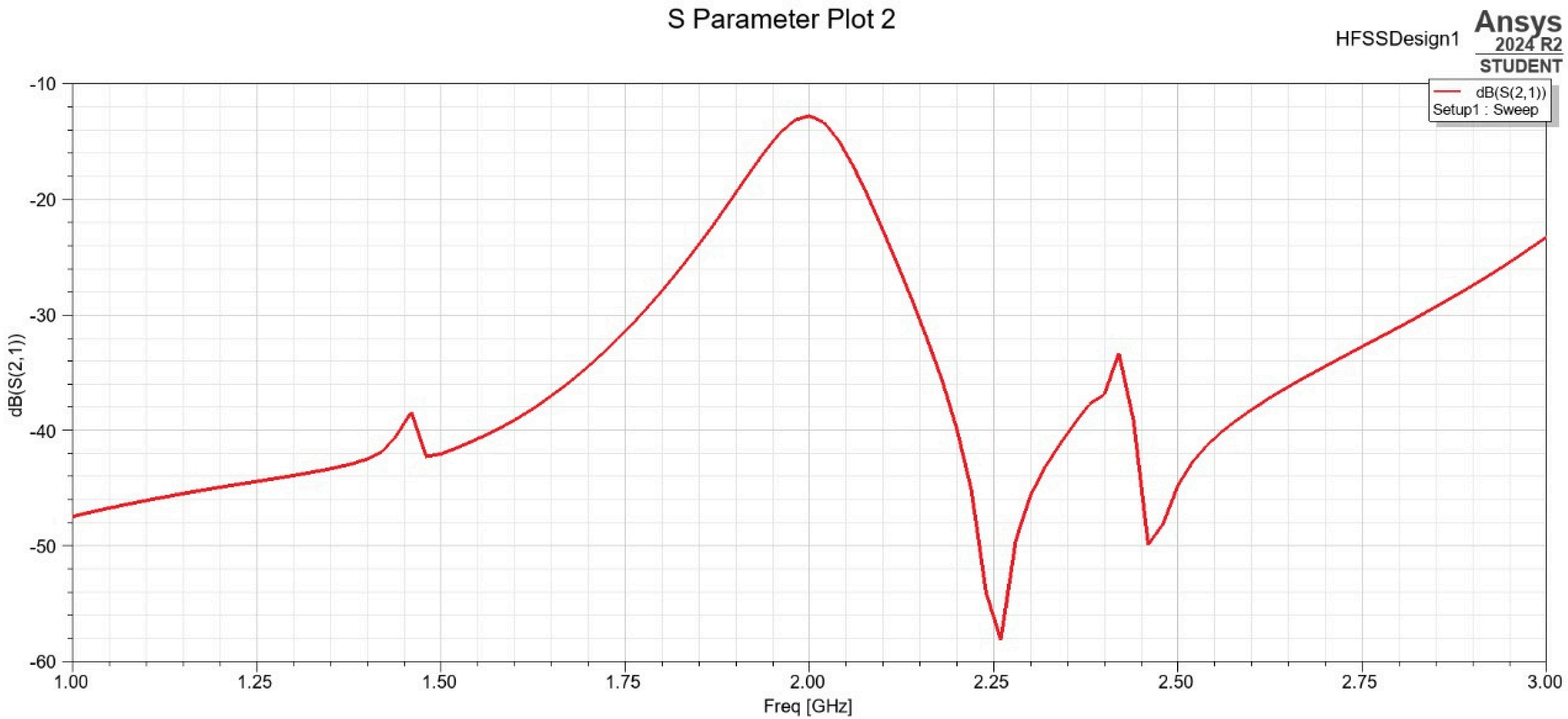

S Parameter Plot 4
HFSSDesign1
Ansys
2024 R2
STUDENT
dB(S(2,2))
Setup1 : Sweep
0
-2
-4
-6
-8
-10
dB(S(2,2))
1.00
1.25
1.50
1.75
2.00
2.25
2.50
2.75
3.00
Freq [GHz]

S12 Plot


S Parameter Plot 3
HFSSDesign1
Ansys
2024 R2
STUDENT
dB(S(1,2))
Setup1 : Sweep
-10
-20
-30
-40
-50
-60
dB(S(1,2))
1.00
1.25
1.50
1.75
2.00
2.25
2.50
2.75
3.00
Freq [GHz]

RT/duroid 5880

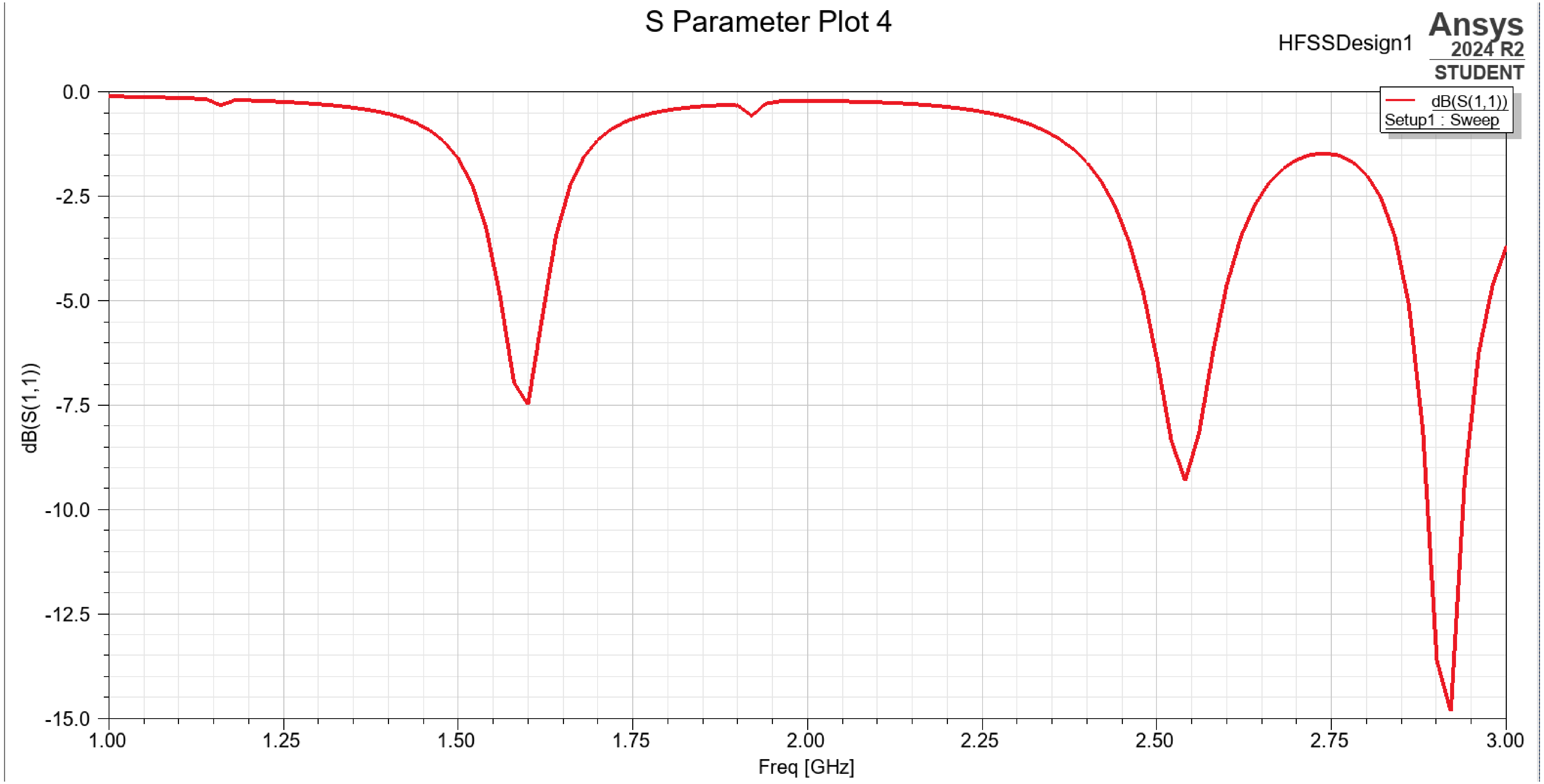
S21 Plot



Rogers RO4003C

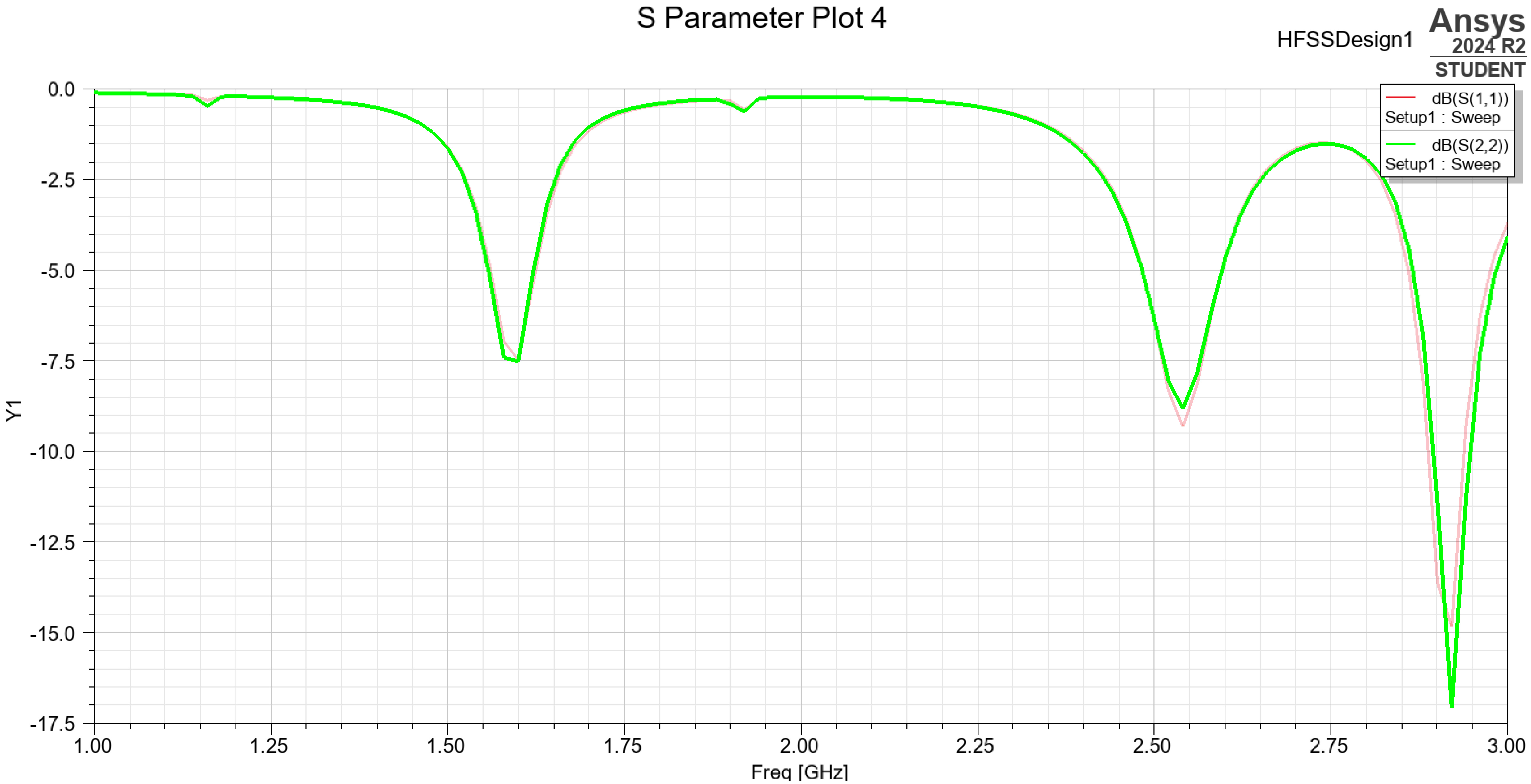
Go, Change the World

S11 Plot



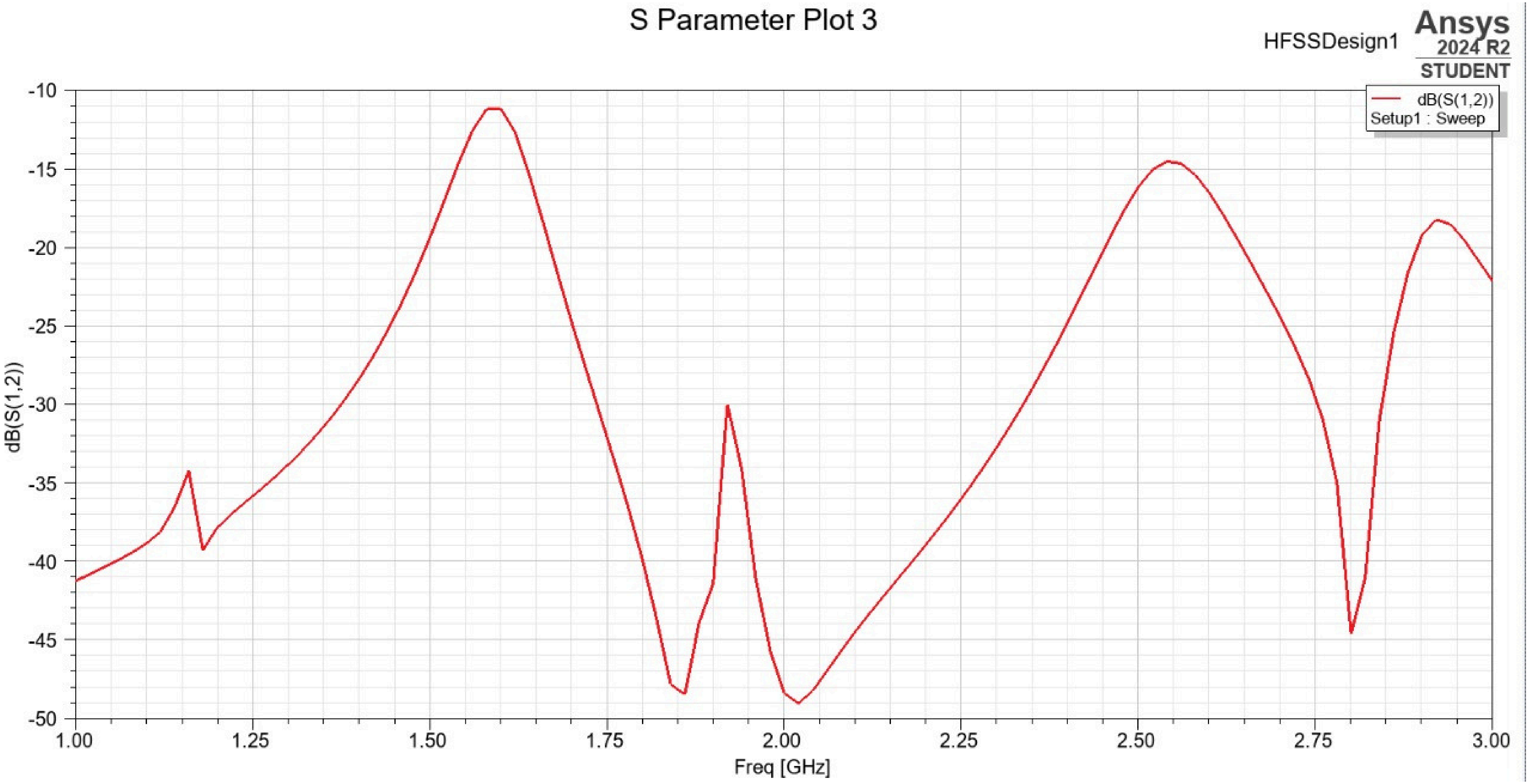
Rogers RO4003C

S22 Plot



Rogers RO4003C

S12 Plot



Rogers RO4003C

S21 Plot